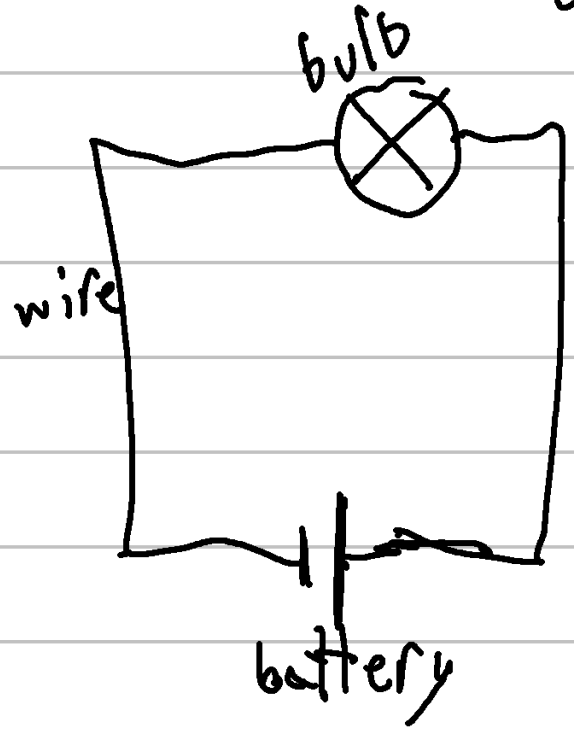
- electricity needs a closed loop to flow if there is any gap or break, the circuit is open and the current stops

- a basic circuit has 3 parts: a power source (battery), a conductor (wire), a load (bulb).

- when you connect all parts with no gap electrons flow from the battery through the wire to the bulb and back - this lights the bulb.



CONDUCTORS AND INSULATORS

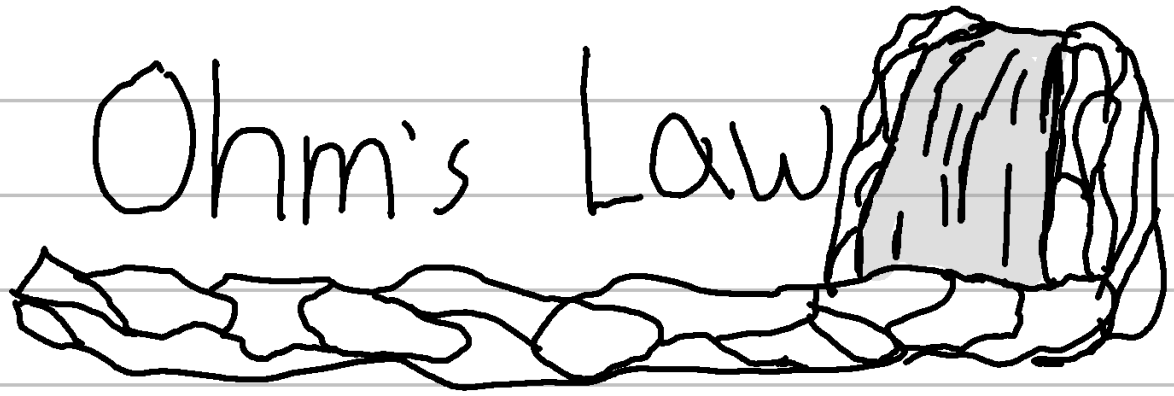
- paperclip - conductor
- coin - conductor
- pencil(graphite)(sharp on both sides - conductor (slow current)
- eraser - insulator
- dollar bill - insulator

- conductors let electrons move freely.
- insulators block to flow of electrons.
- metals are almost always conductors.
- graphite is a non-metal conductor, its a special exception.

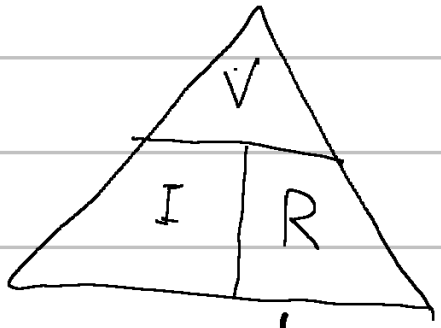
VOLTAGE, current, & RESISTORS

(Ω) Resistance	(V) VOLTAGE	(A) current
10	9	.9
25	9	.36
42	9	.21
107	9	.08

Ohm's Law



$$V = I \cdot R$$

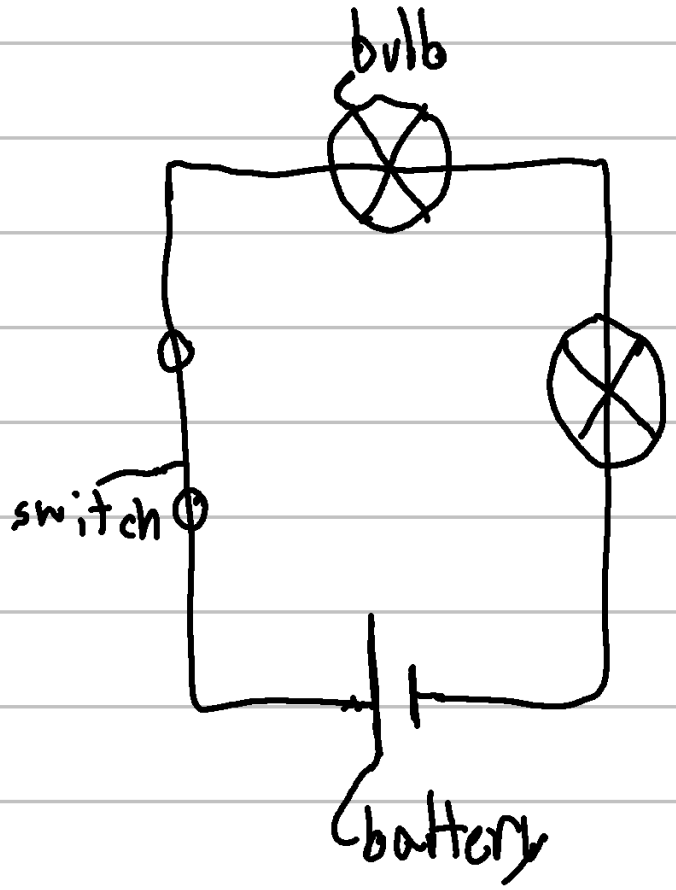
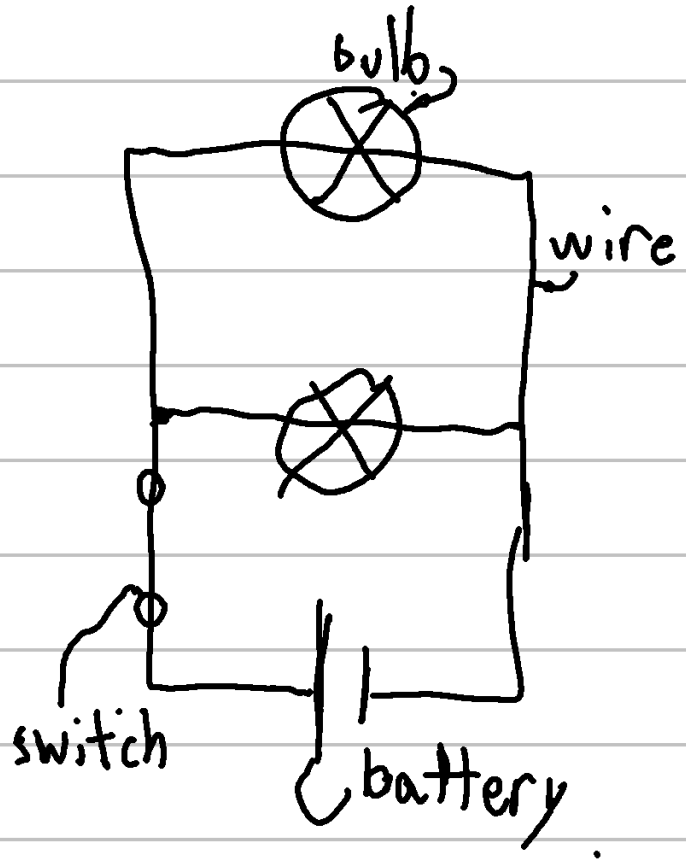


Voltage = current × resistance.

This means: the more resistance in a circuit, the less current will flow (if voltage stays the same)

- a voltmeter measures voltage, it is connected in parallel (across) a component
- an ammeter measures current, it is connected in series (in line) with a component.
- adding more resistance to a circuit causes the current to decrease.
- increasing the voltage causes the current to increase.

SERIES VS PARALLEL

SERIES	PARALLEL
1 Path.	Multiple paths.
1 bulb goes the rest out.	1 bulb goes out the rest stay on.
adding a bulb make them go dimmer	adding a bulb does not make them dimmer
they distribute the voltage.	each get the full voltage

SERIES VS PARALLEL